Giacomo Vianello

Curriculum Vitae

Venice, Italy | Mail | LinkedIn | GitHub

Who I am

I'm Giacomo Vianello and I'm currently enrolled in the last year of my master degree in Control System Engineering at University of Padua, Italy.

Interests: controls (especially optimal and model predictive), robotics and autonomous agents (manipulators and aerial vehicles), learning methods for robotics (Reinforcement Learning), large and vision language models.

Education

Master Degree in Control System Engineering 2026 (Expected)
Institution: University of Padua
Department: Department of Information Engineering (DEI)
Advisor: Ruggero Carli (UniPD)
Final Project: TBA
Final Grade: TBA

Bachelor Degree in Electronic Engineering 2024
Institution: University of Padua
Department: Department of Information Engineering (DEI)
Advisor: Mattia Zorzi (UniPD)
Thesis: "Theory and applications of Kalman Filter to tracking problems."
Final Grade: 110/110

High School Diploma 2021
Institution: Carlo Zuccante, Mestre, VE, Italy
Department: Industrial Technical Institute (ITIS)
Final Project: "Automated high-bay warehouse with 3 DOF manipulator and conveyor belt."
Final Grade: 100/100

Publications & Research

Master's Thesis (In Progress)

G. Vianello, "Title TBA," *Expected 2026*.
Focus: Integration of Vision Language Models and autonomous aerial vehicles for object detection and efficient navigation.

Bachelor's Thesis

G. Vianello, "Theory and application of Kalman Filter to tracking problems," *2024*.
Digital repository: hdl.handle.net/20.500.12608/67440 (Italian).

Awards & Scholarships

Erasmus+ for Traineeship Grant 2025
Awarded competitive funding for an international internship program focused on control systems, robotics and artificial intelligence (3 % acceptance rate).

Academic Merit Scholarship (University of Padua) 2023 – 2025
Awarded for two years to the top 5% of students based on GPA and ECTS credits earned.

Experiences Abroad

Research Intern | Robotics & Artificial Intelligence 2026
Aragon Institute for Engineering Research (I3A), Zaragoza, Spain
Developed and integrated embedded AI algorithms into Micro Aerial Vehicles (MAVs). Focused on real-time object detection, scene understanding, and the design of optimal trajectory planning and control laws.

- *Keywords:* Computer Vision, Embedded Systems, Optimal Control, Robotics.

St. Dunstan's College 2019
London, United Kingdom
Participated in the MOVE Project, an intensive 14-day English language and cultural immersion program.

Professional Experience

Apprentice Electrician Summers 2019 – 2023
DVS Tecnoimpianti, Venice, Italy
Gained hands-on experience in the installation and commissioning of industrial and civil electrical systems.

- Wired and tested complex electrical panels and industrial distribution boards.
- Configured and integrated security architectures, including IP-based video surveillance, fire detection, and intrusion systems.
- Developed troubleshooting skills in electrical circuitry and field components.

Web Developer & IT Maintainer 2023 – Present
DVS Tecnoimpianti, Venice, Italy
Designed and developed the corporate web presence to improve customer engagement and service presentation.

- Developed a responsive professional website using HTML5, CSS3, and JavaScript within a WordPress/Elementor ecosystem.
- Manage ongoing maintenance, SEO optimization, and web security updates.

Skills

Programming Languages: C/C++, Java, Python, Linux Shell, Matlab, Arduino, VHDL, Ladder.

Programming Frameworks/Libraries: ROS 2, Git, PyTorch, LaTeX, Simulink, LabView, Siemens (Simatic and TIA portal), LT Spice.

Spoken Languages: Italian (native), English (fluent), Spanish (basic).

Personal Interests

I'm passionate about Themed Entertainment, where engineering meets immersive storytelling. I extend this into digital work through music production (FL Studio, Ableton) and 3D modeling (Autodesk Fusion, Blender), using these tools to turn abstract ideas into prototypes. My skills in web development and Adobe Creative Suite let me manage projects from initial design to final digital implementation. I also play basketball and lift weights to maintain focus, discipline, teamwork, and steady personal growth.